

Automate e-commerce website Users don't know which products suit their skin/hair type. Use Case User answers a quiz (skin type, concerns, budget) System recommends curated products Filters refine choices (ingredients, brands, price) Value Reduces decision fatigue Increases conversion rates Example A user with acne-prone skin gets a full routine: cleanser + serum + sunscreen. Users struggle to find the right product among millions of options. Use Case Search bar with keywords Filters (price, ratings, brand, delivery speed) AI-based recommendations ("Customers also bought") Value Faster Decision-making Higher product visibility Example User searches "wireless headphones" → filters under ₹2000 → sees top-rated options. */

Code:

```
package sdet_selenium_day_25;
```

```
import java.time.Duration;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
```

```
import org.openqa.selenium.chrome.ChromeDriver;
```

```
import org.testng.Assert;
```

```
import org.testng.annotations.AfterClass;
```

```
import org.testng.annotations.BeforeClass;
```

```
import org.testng.annotations.Test;
```

```
public class Ecommerce_Automation_TestNG {
```

```
    WebDriver driver;
```

```
    @BeforeClass
```

```
    public void setup() {
```

```
        driver = new ChromeDriver();
```

```

driver.manage().window().maximize();

driver.manage().timeouts()
    .implicitlyWait(Duration.ofSeconds(10));

driver.get("https://demowebshop.tricentis.com/");

System.out.println("E-Commerce Website Opened");
}

// =====
// PRODUCT SEARCH
// =====

@Test(priority = 1)
public void productSearch() throws InterruptedException {

    WebElement searchBox = driver.findElement(
        By.id("small-searchterms"));

    searchBox.sendKeys("Laptop");

    driver.findElement(
        By.cssSelector("input[value='Search']"))
        .click();

    Thread.sleep(2000);

    String pageTitle = driver.getTitle();

```

```
System.out.println("Search Page Title : " + pageTitle);

Assert.assertTrue(pageTitle.contains("Search"));

System.out.println("Product Search Successful");
}
```

```
// =====
// APPLY FILTERS
// =====
```

```
@Test(priority = 2)
public void applyFilters() throws InterruptedException {
```

```
    // Example category click
```

```
    driver.findElement(
        By.linkText("Electronics"))
        .click();
```

```
    Thread.sleep(2000);
```

```
    System.out.println("Electronics Category Opened");
```

```
    // Open Cell Phones
```

```
    driver.findElement(
        By.linkText("Cell phones"))
```

```

        .click();

Thread.sleep(2000);

System.out.println("Filters Applied Successfully");
}

// =====
// AI PRODUCT RECOMMENDATION
// =====

@Test(priority = 3)
public void recommendedProducts() {

    WebElement recommendedProduct = driver.findElement(
        By.linkText("Smartphone"));

    String productName = recommendedProduct.getText();

    System.out.println("Recommended Product : "
        + productName);

    Assert.assertTrue(recommendedProduct.isDisplayed());

    System.out.println("AI Recommendation Displayed");
}

// =====
// QUIZ BASED PRODUCT SELECTION

```

```
// =====

@Test(priority = 4)
public void skinCareQuizSimulation()
    throws InterruptedException {

    System.out.println("Skin Type : Acne-Prone");

    System.out.println("Concern : Pimples");

    System.out.println("Budget : ₹2000");

    // Simulated recommendation

    String recommendedRoutine =
        "Cleanser + Serum + Sunscreen";

    System.out.println("Recommended Routine : "
        + recommendedRoutine);

    Assert.assertEquals(recommendedRoutine,
        "Cleanser + Serum + Sunscreen");

    System.out.println("Quiz Recommendation Successful");

    Thread.sleep(2000);
}

// =====
```

```
// ADD PRODUCT TO CART

// =====

@Test(priority = 5)
public void addToCart() throws InterruptedException {

    driver.findElement(
        By.linkText("Smartphone"))
        .click();

    Thread.sleep(2000);

    driver.findElement(
        By.id("add-to-cart-button-43"))
        .click();

    Thread.sleep(3000);

    WebElement successMessage = driver.findElement(
        By.cssSelector(".content"));

    String message = successMessage.getText();

    System.out.println("Cart Message : "
        + message);

    Assert.assertTrue(message.contains(
        "The product has been added"));
```

```
        System.out.println("Product Added To Cart");  
    }
```

```
@AfterClass
```

```
public void tearDown() {
```

```
    driver.quit();
```

```
        System.out.println("Browser Closed");
```

```
    }
```

```
}
```